

우주비행학

Home Work 5

Due date: June 5, 2018

- 1) Problem 1 in Chapter 10
- 2) Problem 2 in Chapter 10
- 3) Prove that the L_2 point is an unstable equilibrium point in the restricted three-body problem.

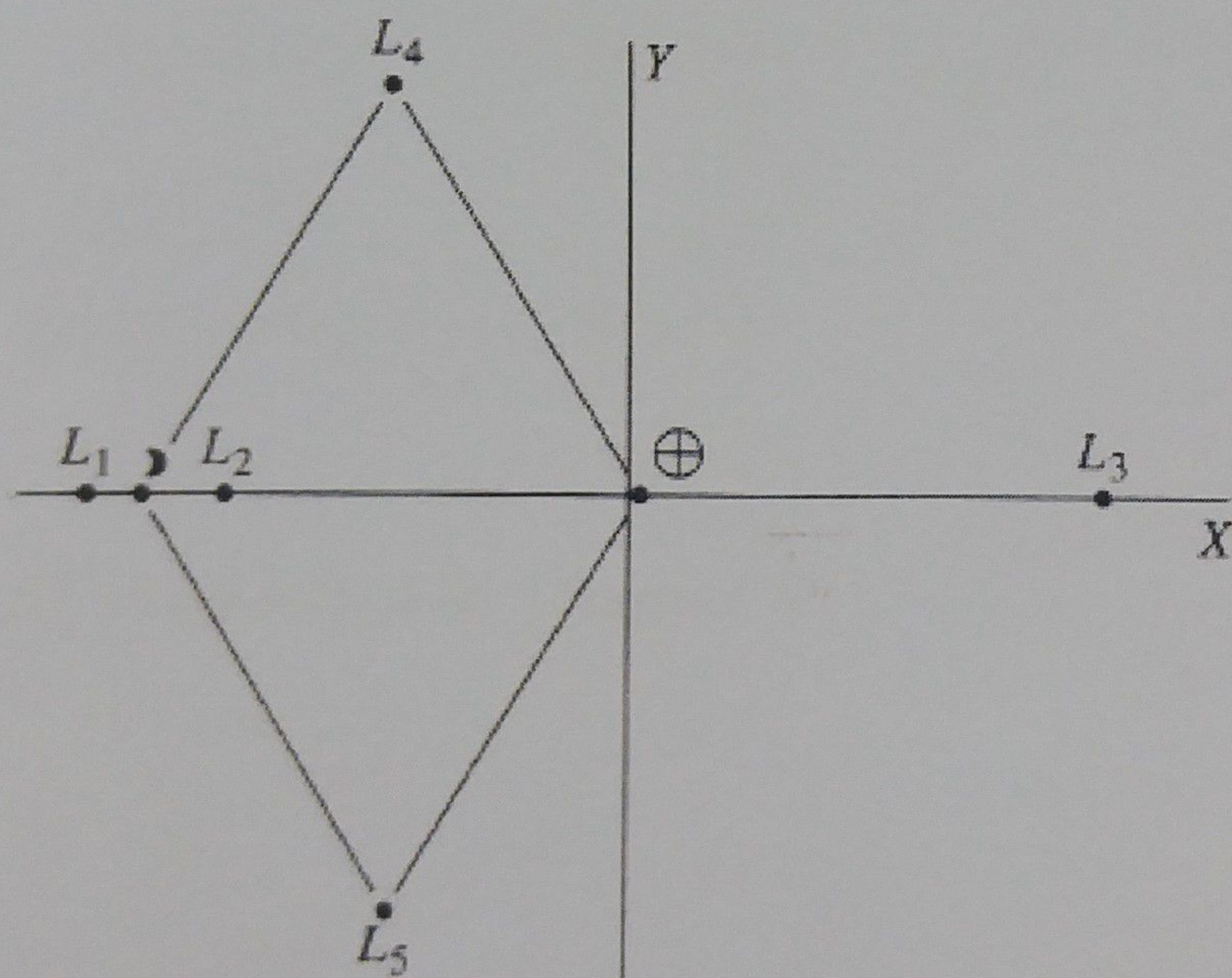


FIGURE 10.2
Equilibrium points for the restricted problem.

10.8

REFERENCES AND FURTHER READING

It is not necessary to cite any reference for the restricted problem beyond Szebehely's magnificent book. The study of periodic orbits and surface of section techniques are both recent additions to the arsenal of dynamical techniques and can be applied to any other nonlinear system, as well. However, the study of chaos, in particular, is too new to cite an introductory book.

Szebehely, V. *Theory of Orbits*, Academic Press, New York, 1967.

10.9

PROBLEMS

1. Use the Jacobi integral to calculate the speed needed to depart from the vicinity of the earth ($r_1 = 1/60$ distance units) to arrive at L_2 . The value of C at L_2 is -1.59411 . (Ignore small terms in the expression for Jacobi's integral.)
2. Determine the minimum speed needed departing earth orbit to arrive at L_4 .
3. Verify that L_2 is located near $x = -0.83702$. Calculate the λ values at L_2 and the eigenvectors $\mathbf{A}$ for the case of the earth-moon system, $\mu = 0.01213$. Verify that this equilibrium point is unstable.
4. Show that the inertial velocity components $\dot{X}$ and $\dot{Y}$ (still resolved along the rotating axes) are given by

$$\dot{X} = \dot{x} - y, \quad \dot{Y} = \dot{y} + x \quad (10.38)$$

Show that Jacobi's integral takes the form

$$C = \frac{1}{2}(\dot{X}^2 + \dot{Y}^2) + \dot{X}y - \dot{Y}x - \frac{1-\mu}{r_1} - \frac{\mu}{r_2} \quad (10.39)$$

When the third object is far from the smaller mass, and ignoring terms of order μ , show that this becomes

$$C \approx \mathcal{E} - H \approx -\frac{1}{2a} - \sqrt{a(1-e^2)} \quad (10.40)$$

in terms of the two-body elements. This is *Tisserand's criterion* for the identification of comets after they have undergone a close approach to Jupiter. In consulting Chapter 2, remember that μ has a completely different meaning in the two-body problem.

5. Verify the form of the linearized equations for δx and δy (10.22) and (10.23) near an equilibrium point. Also, derive the vertical equation for δz and show it always has the form of a harmonic oscillator and is therefore always stable.
6. Design a feedback controller for a satellite which is to stationkeep in the vicinity of L_2 . (Hint: Consider feeding back the amplitudes of the relative position and velocity projected on the "outgoing" unstable eigenvector $\mathbf{A}$; the eigenvector associated with a positive real λ value.)